

Skills Summary

Differentiating Implicit Relations

Use this reference while completing your worksheet or when studying.

Skill 1 — Differentiating a Relation Implicitly

Why it is legal: on any arc where y is a differentiable function of x , both sides of a relation are the same function of x , so both sides may be differentiated with respect to x .

The bookkeeping: every y that gets differentiated leaves behind a factor $\frac{dy}{dx}$ (chain rule).

Term by term: $x^3 \rightarrow 3x^2$; $y^3 \rightarrow 3y^2 \frac{dy}{dx}$; $xy \rightarrow y + x \frac{dy}{dx}$; $x^2y \rightarrow 2xy + x^2 \frac{dy}{dx}$; a constant $\rightarrow 0$.

Then solve: collect the $\frac{dy}{dx}$ terms on one side, factor, divide. The answer usually contains both x and y , and that is correct.

Example: Find $\frac{dy}{dx}$ for $x^2 + y^2 = 9$.

Step 1. The relation cannot be solved for y as a single function, but it can be differentiated as it stands, which is exactly what implicit differentiation is for.

Step 2. Differentiating gives $2x + 2y \frac{dy}{dx} = 0$, so $2y \frac{dy}{dx} = -2x$ and $\frac{dy}{dx} = -\frac{x}{y} = \frac{-x}{\sqrt{9-x^2}}$ on the upper arc.

Try it: Find $\frac{dy}{dx}$ for $x^2 + y^2 = 16$, and write it in terms of x alone on the upper arc.

$\frac{-x}{\sqrt{16-x^2}} = \frac{-x}{4}$ check

Skill 2 — The Slope at a Point on the Curve

An implicit derivative needs both coordinates. $\frac{dy}{dx}$ comes out in terms of x and y , so a slope cannot be read from an x -value alone.

Check the point first. Substitute it into the original relation. If the two sides do not agree, the point is not on the curve and any slope computed there is meaningless.

Then substitute both coordinates into the derivative and simplify.

Example: For $x^2y = 12$, find the slope at $(2, 3)$.

Step 1. Before differentiating anything, confirm the point is on the curve: $2^2 \cdot 3 = 12$, so it is, and the slope there will mean something.

Step 2. The product rule gives $2xy + x^2 \frac{dy}{dx} = 0$, so $\frac{dy}{dx} = -\frac{2y}{x}$. At $(2, 3)$ that is -3 .

Try it: For $x^2y = 18$, check that $(3, 2)$ is on the curve and find the slope there.

check: $\frac{8}{4}$

Skill 3 — Tangent Lines and Special Points

Tangent line: slope m from the implicit derivative at the point, then $y - y_1 = m(x - x_1)$ through that point.

Horizontal tangent: the numerator of $\frac{dy}{dx}$ is zero and the denominator is not. Set the numerator to 0, substitute the resulting relation back into the original equation, and solve for the points.

Vertical tangent: the denominator is zero and the numerator is not — the slope is undefined there, and y is not a function of x near that point.

Example: Write the tangent to $x^2 + y^2 = 10$ at $(1, 3)$.

Step 1. A tangent line needs a point and a slope, and the point is given, so the only missing piece is the implicit derivative evaluated there.

Step 2. From $2x + 2y \frac{dy}{dx} = 0$, $\frac{dy}{dx} = -\frac{x}{y} = -\frac{1}{3}$ at $(1, 3)$, giving the line $y = -\frac{1}{3}(x - 1) + 3$.

Try it: Write the tangent to $x^2 + y^2 = 20$ at $(2, 4)$.

check: $4 + (4 - x)\frac{2}{1} = 8$

(!) Watch out: differentiating y^2 as $2y$ is the classic slip — the chain-rule factor $\frac{dy}{dx}$ is not optional. And when a term holds both variables, such as xy , the product rule comes first and the chain rule applies inside it.